

The Casson Invariant

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1 Introduction

The Casson invariant is a $\mathbb{Z}$ -valued invariant of integer homology 3-spheres ($\mathbb{Z}HS^3$'s). Very roughly, it counts inequivalent irreducible $SU(2)$ -representations of $\pi_1 Y$. In this article we will give its important properties, detail its precise construction, and say a few words about the proof that it is an invariant. We will not, however, show that the given construction satisfies the claimed properties.

This writeup is based heavily on Saveliev's *Lectures on the Topology of 3-Manifolds: An Introduction to the Casson Invariant* [Sav12].

2 Properties

The Casson invariant is some sense a lift of the *Rokhlin invariant*, which we now recall.

Theorem 1 (Rokhlin's theorem). Let W be a closed smooth 4-manifold with even intersection form. Let $\sigma(W)$ denote the signature of this form. Then, $\sigma(W) \equiv 0 \pmod{16}$.

Since every integer homology sphere Y is obtained from integral surgery on a link, Y bounds a 2-handlebody whose attaching circles and their framings are given by the link components and surgery coefficients of Y . In fact, any integer homology sphere bound a 2-handlebody W with *even* intersection form. Even intersection forms satisfy $\sigma(W) \equiv 0 \pmod{8}$. By Rokhlin's theorem, whether $\sigma(W)$ is 0 or 8 mod 16 depends only on Y . This allows for the definition of the following invariant.

Definition 1 (Rokhlin invariant). Let Y be an $\mathbb{Z}HS^3$. Let W be any smooth 4-manifold with even intersection form and $\partial W = Y$. Then, we define the *Rokhlin invariant* of Y to be

$$\mu(Y) := \frac{\sigma(W)}{8} \pmod{2}.$$

We now list properties the yet-to-be-defined Casson invariant λ will satisfy.

1. $\lambda(Y_1 \# Y_2) = \lambda(Y_1) + \lambda(Y_2)$
2. Let $\bar{Y}$ denote Y with reversed orientation. Then, $\lambda(\bar{Y}) = -\lambda(Y)$. In particular, λ vanishes whenever Y admits an orientation-reversing self-diffeomorphism.
3. Let $k \subset Y$ be a knot, and let $Y + \frac{p}{q}k$ denote p/q surgery along k . Then, the quantity

$$\lambda'(k) := \lambda\left(Y + \frac{1}{m+1}k\right) - \lambda\left(Y + \frac{1}{m}k\right)$$

is well defined, i.e. the righthand side is independent of m . In particular, for $k \subseteq S^3$, we are defining $\lambda'(k) = \lambda(S^3 + k)$.

4. For any knot $k \subseteq Y$, $\lambda'(k) = \frac{1}{2}\Delta_k''(1)$, where Δ_k'' is the formal second derivative of the Alexander polynomial of k . In particular, $\lambda'(T_{2,3}) = 1$.
5. $\lambda(Y) \equiv \mu(Y) \pmod{2}$.

3 Background

The Casson invariant is defined using $SU(2)$ -representation spaces. We now recall the corresponding definitions.

Definition 2 ($SU(2)$). The *special unitary group* is the subgroup

$$SU(2) = \{A \in SL_2(\mathbb{C}) \mid AA^\dagger = I_2\},$$

where $\dagger$ denotes conjugate-transpose. The group $SU(2)$ admits the following more explicit description:

$$SU(2) = \left\{ \begin{bmatrix} a & b \\ -\bar{b} & \bar{a} \end{bmatrix} : a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\},$$

which may also be taken as a definition.

The second definition gives an embedding $SU(2) \hookrightarrow \mathbb{C}^2$, which in fact can be seen to identify $SU(2) \cong S^3$. This isomorphism holds at the level of lie groups, where S^3 has group structure given by quaternion multiplication.

Definition 3 (Representation, $SU(2)$ -representation). A *representation* (V, ϕ) of a group G is an action of G on V , equivalently a group homomorphism $\phi : G \rightarrow GL(V)$. An *$SU(2)$ -representation* is an action of G on $\mathbb{C}^2$ via unitary matrices, i.e. a group homomorphism $\phi : G \rightarrow SU(2) \subseteq GL_2(\mathbb{C})$.

Definition 4 (Irreducible representation). A representation $\phi : G \rightarrow \mathrm{GL}(V)$ is called *reducible* if there is a non-0 proper subspace $W \subseteq V$ preserved by G , i.e. $\phi(g)(W) \subseteq W$ for all $g \in G$ (the containment will always be an equality if W is finite-dimensional). If a representation is not reducible, we call it *irreducible*.

Remark 1. If an $\mathrm{SU}(2)$ representation $\phi : G \rightarrow \mathrm{SU}(2)$ is reducible, then it must preserve some one-dimensional subspace W . Taking a spanning unit vector for W and an orthogonal unit vector as a basis, we conclude that $\phi(g)$ must be upper-triangular for all g . Using our matrix entry description of $\mathrm{SU}(2)$, this forces the top right entry to be 0 as well. Thus,

$$\phi(g) = \begin{bmatrix} e^{i\theta(g)} & 0 \\ 0 & e^{-i\theta(g)} \end{bmatrix}$$

for some function $\theta : G \rightarrow S^1$. Notably, for $\mathrm{SU}(2)$ representations, reducibility is equivalent to abelian image.

Definition 5 (Equivalence of representations). Two representations $\phi, \psi : G \rightarrow \mathrm{GL}(V)$ are *equivalent* if they are related by a change of basis, i.e. there is some $P \in \mathrm{GL}(V)$ such that $\phi(g) = A\psi(g)A^{-1}$ for all A .

Remark 2. It turns out that whenever two $\mathrm{SU}(2)$ representations are equivalent, the equivalence is realized by a unitary change of basis matrix $A \in \mathrm{SU}(2)$.

Definition 6 (($\mathrm{SU}(2)$)-Representation space). Let G be a group. The set of all representations $\phi : G \rightarrow \mathrm{SU}(2)$ is called the *representation space* $R(G)$ of G . Endowing G with the discrete topology, we upgrade $R(G)$ from a set to a space by giving it the compact-open topology. We similarly let $R^{\mathrm{irr}}(G)$ denote the space of irreducible representations of G .

For M a manifold, we respectively use $R(M)$ and $R^{\mathrm{irr}}(M)$ to denote the representation spaces $R(\pi_1 M)$ and $R^{\mathrm{irr}}(\pi_1 M)$ of the fundamental group.

Remark 3. For $G = \langle x_1, \dots, x_n \mid \text{relations} \rangle$ a finitely generated group, there is a much more hands-on description of the topology of $R(G)$. Specifically, a representation ϕ is determined by the images $\phi(x_1), \dots, \phi(x_n)$ of the generators. The map

$$\begin{aligned} R(G) &\longrightarrow \mathrm{SU}(2)^n \\ \phi &\longmapsto (\phi(x_1), \dots, \phi(x_n)) \end{aligned}$$

gives an embedding $R(G) \hookrightarrow \mathrm{SU}(2)^n$, and the corresponding subspace topology agrees with the compact-open topology. If G is finitely presented, then $R(G)$ inherits the structure of a real-algebraic set in $\mathbb{R}^{4n}$, since the relations cut G out of $\mathrm{SU}(2)^n \cong (S^3)^n \subset (\mathbb{R}^4)^n$ via polynomial equations.

Definition 7 (Character variety, reduced representation space). Let G be a group, and $R(G)$ its representation space. The group $\mathrm{SU}(2)$ acts on $R(G)$ by $A(\phi)(g) = A\phi(g)A^{-1}$. By Remark 2, the orbits of this action correspond exactly

to equivalence classes of representations. The action of $SU(2)$ is unfaithful, with kernel $Z(SU(2)) = \{\pm 1\}$. Thus, there is a faithful action of $SU(2)/Z(SU(2)) \cong SO(3)$ on $R(G)$. We call the quotient $R(G)/SO(3)$ the *character variety* of G and denote it $\chi(G)$. Restricting to irreducible representations, we also define the *reduced representation space* $\mathcal{R}(G) := R^{\text{irr}}(G)/SO(3)$.

Remark 4. Since $SO(3)$ is compact, $\chi(G)$ is Hausdorff. This is one reason why we work specifically with $SU(2)$ representations instead of, say, all representations on $\mathbb{C}^2$.

Remark 5. The action of $SO(3)$ on $R(G)$ is faithful, but not free. Thus, even if $R(G)$ is a smooth manifold, the character variety will be singular at the representations with nontrivial stabilizer. However, the elements with nontrivial stabilizer are exactly the reducible representations. Thus, if we restrict to *irreducible* representations, then we do get a free action. In other words, as long as $R^{\text{irr}}(G)$ is a manifold, $\mathcal{R}(G)$ will be a manifold of dimension 3 less. This is arguably the main motivation for defining $\mathcal{R}(G)$.

Remark 6 (Functoriality). Let $\psi : G \rightarrow H$ be any group homomorphism. The homomorphism ψ induces a continuous map $\psi^* : R(S) \rightarrow R(G)$ via $\phi \mapsto \phi \circ \psi$. If ψ is surjective, then ψ^* is injective. This will be important later.

Definition 8 (Zariski tangent space). Let $X \subseteq M$ be a real-algebraic subset of $\mathbb{R}^n$. The set X is cut out by some polynomials $f_1, \dots, f_r$, meaning

$$X = \{x \in \mathbb{R}^n : f_1(x) = \dots = f_r(x) = 0\}.$$

There are in general many possible choices for the $\{f_i\}$; choose a set so that the ideal $(f_1, \dots, f_r) \subseteq \mathbb{R}[x_1, \dots, x_n]$ is radical. Let $\ell_1, \dots, \ell_r$ denote the linear approximations of the $\{f_i\}$ about p . The *Zariski tangent space* $T_p X$ of X at p is the hyperplane cut out by $\ell_1, \dots, \ell_r$.

Example 1. If X is locally a smooth submanifold near p , then the Zariski tangent space is just the usual tangent space.

Example 2. Let $X = \{(t^2, t^3) : t \in \mathbb{R}\}$. X is cut out by $f(x, y) = y^2 - x^3$. Let $p = (x_0, y_0) \in X$. Then, we may write

$$f(x, y) = (y - y_0)^2 + 2y_0(y - y_0) + y_0^2 + (x - x_0)^3 + 3x_0(x - x_0)^2 + 3x_0^2(x - x_0) + x_0^3.$$

We can then see that the linear approximation of f at p is $\ell_p(x, y) = 2y_0(y - y_0) + 3x_0^2(x - x_0)$. For $p \neq (0, 0)$, the condition $\ell_p(x, y) = 0$ cuts out the line tangent to X at p . For $p = (0, 0)$, we get $\ell_p \equiv 0$, so ℓ_p always vanishes and thus the Zariski tangent space at 0 is the whole plane. This jump in tangent space dimension (from 1 to 2) corresponds to the fact that the graph of $y^2 = x^3$ has a cusp at the origin.

Remark 7. Notice that in these examples, the Zariski tangent space dimension is always at least the dimension of X , and is higher at singularities. This is true in general. (What do I mean by $\dim X$? There are many equivalent ways to define it. Hausdorff dimension as a subset of $\mathbb{R}^n$ is one such way.)

Remark 8 (Don't get hung up on this paragraph; you need only understand the last 16 words.). Let M be a manifold, with $\pi_1 M$ finitely presented. If one directly computes $T_\phi R(M)$, then one gets $H^1(\pi_1 M, \mathfrak{su}(2))$ in group cohomology, where the action of $\pi_1 M$ on $\mathfrak{su}(2)$ is the composition of ϕ and the adjoint representation $\text{Ad} : \text{SU}(2) \rightarrow \text{Aut}(\mathfrak{su}(2))$. Let ϵ denote the trivial representation. With some further yoga, one can convert from group cohomology to bona fide singular cohomology, yielding a natural isomorphism $T_\epsilon R(M) \cong H^1(M, \mathfrak{su}(2))$. Concretely, you can just think of $\mathfrak{su}(2)$ as $\mathbb{R}^3$.

4 Construction

Let $Y = M_1 \cup_\Sigma M_2$ be a Heegaard splitting of a $\mathbb{Z}HS^3$ into 2-handlebodies M_1 and M_2 glued along their common genus g boundary Σ . The splitting is summarized in the following diagram.

$$\begin{array}{ccccc} & & M_1 & & \\ \nearrow i_1 & & & \nwarrow j_1 & \\ \Sigma & & & & Y \\ \searrow i_2 & & M_2 & \swarrow j_2 & \end{array}$$

Since M_1 and M_2 are 4-balls with g handles attached each, we see $M_1 = \langle \alpha_1, \dots, \alpha_g \rangle$ and $M_2 = \langle \beta_1, \dots, \beta_g \rangle$ for some curves $\alpha_1, \dots, \alpha_g, \beta_1, \dots, \beta_g \subseteq \Sigma$. Thus, the induced maps $i_{1,*}$ and $i_{2,*}$ on π_1 are surjective. By Seifert-Van-Kampen, $j_{1,*}$ and $j_{2,*}$ are as well. Applying this to the above diagram, we get

$$\begin{array}{ccccc} & & \pi_1 M_1 & & \\ \nearrow i_{1,*} & & & \nwarrow j_{1,*} & \\ \pi_1 \Sigma & & & & \pi_1 Y \\ \searrow i_{2,*} & & \pi_1 M_2 & \swarrow j_{2,*} & \end{array}$$

Now take representation spaces to yield

$$\begin{array}{ccccc} & & R(M_1) & & \\ \nearrow i_1^* & & & \nwarrow j_1^* & \\ R(\Sigma) & & & & R(Y) \\ \searrow i_2^* & & R(M_2) & \swarrow j_2^* & \end{array}$$

For the rest of this section we will be a little sloppy, and identify each representation space with its image in $R(\Sigma)$. Our goal is to construct an invariant by taking the intersection number of $R(M_1)$ and $R(M_2)$ inside $R(\Sigma)$, the idea being that this should measure the representations coming from Y , since all the squares are Cartesian. However, this won't work as stated. Each of M_1 and M_2 have free fundamental groups, so their representation spaces are each diffeomorphic to $\text{SU}(2)^g$, and in particular are $3g$ -dimensional. On the other hand, $\pi_1 \Sigma$ is generated by $2g$ elements with one relation, and so has representation

space homeomorphic to $\mathrm{SU}(2)^{2g-1}$, which has dimension $6g - 3$. The assiduous reader will now point out that generic spaces of these dimensions produce a 3-dimensional intersection, which is not good for counting intersection numbers.

However, look what happens when we look at *reduced* representation spaces. Every dimension drops by 3, so we are now looking at two $(3g - 3)$ -dimensional manifolds sitting inside a $(6g - 6)$ -dimensional manifold. If everything were compact smooth manifolds, we would dust our hands off and call it a day. However, each of $\mathcal{R}(\Sigma), \mathcal{R}(M_1), \mathcal{R}(M_2)$ are open manifolds, so we have to do some more work to ensure this intersection number is finite and well-defined. For these steps we will actually need to use the fact that Y is a $\mathbb{Z}HS^3$.

Theorem 2. The intersection $\mathcal{R}(M_1) \cap \mathcal{R}(M_2) \subseteq \mathcal{R}(\Sigma)$ is compact.

Proof. The overarching strategy will be to work upstairs, with $R(\cdot)$, and control the reduced locus as much as possible.

Lemma 1. $R(M_1) \cap R(M_2) = (R^{\mathrm{irr}}(M_1) \cap R^{\mathrm{irr}}(M_2)) \cup \{\epsilon\}$, where ϵ is the trivial representation.

Proof. The manifold Y is an integer homology sphere by hypothesis, so $H_1(Y)$ is trivial. By Remark 1, any reduced $\mathrm{SU}(2)$ representation has abelian image. Therefore, any reduced $\mathrm{SU}(2)$ representation factors through $[\pi_1 Y, \pi_1 Y] = H_1(Y) = 0$ and hence is trivial. $\square$

Lemma 2. The trivial representation ϵ is an isolated point of $R(M_1) \cap R(M_2)$.

Proof. The idea for this proof is to show that $T_\epsilon R(\Sigma) = T_\epsilon R(M_1) \oplus T_\epsilon R(M_2)$. The reader may object to this strategy, asking “If $R(M_1)$ and $R(M_2)$ are $3g$ -dimensional, how can their tangent spaces sum to the tangent space of something $(6g - 3)$ -dimensional?” This is a great question, and if each representation space was a smooth manifold, we would indeed be in trouble. Our saving grace is the fact that by “tangent space”, we mean *Zariski* tangent space, coming from the real-algebraic structure described in Remark 3. With this structure, $R(M_1)$ and $R(M_2)$ are still smooth manifolds of dimension $3g$, but $R(\Sigma)$ is singular at reduced representations, and in particular has Zariski tangent space dimension $6g$ at ϵ .

We note here that the proof of the fact that the singular locus of $R(\Sigma)$ is exactly the locus of reduced representations is a nasty 3 page long calculation, hence why we omit it. We also note that this fact is one reason why it is preferable to take intersections in $\mathcal{R}(\Sigma)$ as opposed to all of $\chi(\Sigma)$.

The reader may now pose a new objection: “The reason we would want a decomposition of tangent spaces is to use the fact that a transverse intersection of smooth manifolds is locally smooth with dimension equal to the dimension of the tangent space overlap. Why is a decomposition of Zariski tangent spaces still sufficient?” To this objection, we provide the perhaps unsatisfying answer: there are similar results for real algebraic sets that we will not prove or even

precisely state here; we leave it as a (reasonable!) exercise to the reader to show that a tangent space decomposition implies isolated intersection point.

We now show that the intersection of $R(M_1)$ and $R(M_2)$ in $R(\Sigma)$ is transverse, using Remark 8, Meyer-Vietoris, and the fact that Y is a $\mathbb{Z}HS^3$. Apply Meyer-Vietoris to $M_1 \cup M_2 = Y$:

$$\begin{aligned} \dots \longrightarrow H^1(Y, \mathfrak{su}(2)) &\longrightarrow H^1(M_1, \mathfrak{su}(2)) \oplus H^1(M_2, \mathfrak{su}(2)) \\ &\xrightarrow{\sim} H^1(\Sigma, \mathfrak{su}(2)) \longrightarrow H^2(Y, \mathfrak{su}(2)) \longrightarrow \dots \end{aligned}$$

Since Y is a $\mathbb{Z}HS^3$, the H^1 and H^2 of Y vanish, yielding an isomorphism between the middle two terms. Recalling from Remark 8 that $H^1(M, \mathfrak{su}(2)) \cong T_\epsilon R(M)$ for each manifold M in the sequence, we get the desired transversality. $\square$

Combining Lemmas 1 and 2, we conclude that $R^{\text{irr}}(M_1) \cap R^{\text{irr}}(M_2)$ is compact. Quotients of compact spaces are compact, so $\mathcal{R}(M_1) \cap \mathcal{R}(M_2)$ is compact, proving Theorem 2. $\square$

Knowing now that $\mathcal{R}(M_1) \cap \mathcal{R}(M_2)$ is compact, we may rejoice in taking the algebraic intersection number $\langle \mathcal{R}(M_1), \mathcal{R}(M_2) \rangle$. Even though $\mathcal{R}(M_1), \mathcal{R}(M_2), \mathcal{R}(\Sigma)$ are open, $\langle \mathcal{R}(M_1), \mathcal{R}(M_2) \rangle$ can be shown to be well-defined using cohomology with compact support.

Provisional Definition (Casson Invariant). We provisionally define the *Casson invariant* $\lambda(Y)$ of an integer homology 3-sphere Y to be

$$\lambda(Y) := \langle \mathcal{R}(M_1), \mathcal{R}(M_2) \rangle.$$

Remark 9. On the level of sets, $\mathcal{R}(Y) = \mathcal{R}(M_1) \cap \mathcal{R}(M_2)$. So, why did we go through all the trouble of taking a Heegaard splitting and doing all this terrible work with Zariski tangent spaces and so on, rather than just taking $\lambda(Y) = \#\mathcal{R}(Y)$? There are two reasons. The first is that $\mathcal{R}(M_1)$ and $\mathcal{R}(M_2)$ are not everywhere-transverse, so $\mathcal{R}(Y)$ need not be a nice, finite set of points. Indeed, it can contain copies of S^2 , among other complications. One might hear that and be inclined to just count connected components of $\mathcal{R}(Y)$. This is better, but will still lack some nice properties we want, because in contrast to $\langle \mathcal{R}(M_1), \mathcal{R}(M_2) \rangle$, $\#\pi_0(\mathcal{R}(\Sigma))$ is an *unsigned* count.

5 Invariance (and the actual Casson invariant)

In defining the Casson invariant, we started by taking a Heegaard splitting, and relied heavily on that choice of splitting. So, there is some work to do in showing well-definedness.

Given any Heegaard Splitting, we can get another splitting by “turning it upside down,” i.e. swapping the roles of M_1 and M_2 , and the α_i and β_i curves. If one runs back through the construction of the Casson invariant, carefully tracking orientations, one comes to the alarming conclusion that this swapping multiplies the “invariant” by $(-1)^g$, where g is the genus of the splitting. This is not a huge issue though, since if we redefine $\lambda(Y) := (-1)^g \langle \mathcal{R}(M_1) \cap \mathcal{R}(M_2) \rangle$, then we still have a candidate for an invariant.

This revised provisional definition will indeed turn out to depend only on Y . It also turns out to always be even. Thus, we provide a final, actual definition of the Casson invariant as follows.

Definition 9 (Casson invariant). The *Casson invariant* $\lambda(Y)$ of an integer homology 3-sphere Y with Heegaard splitting $Y = M_1 \cup_\Sigma M_2$ is the integer

$$\lambda(Y) := \frac{1}{2}(-1)^g \langle \mathcal{R}(M_1), \mathcal{R}(M_2) \rangle,$$

where $\langle \mathcal{R}(M_1), \mathcal{R}(M_2) \rangle$ denotes the algebraic intersection number of $\mathcal{R}(M_1)$ and $\mathcal{R}(M_2)$ inside $\mathcal{R}(\Sigma)$ and g is the genus of Σ .

Like Heegaard-Floer homology, the proof of invariance relies on general theory showing that all Heegaard splittings are related by certain simple moves (à la Reidemeister moves), allowing us to just prove invariance under those moves. We now recall the general theory.

Definition 10 (Equivalence of Heegaard splittings). We say two Heegaard splittings $M_1 \cup_\Sigma M_2 = Y = M'_1 \cup_{\Sigma'} M'_2$ are *equivalent* if there is a self-diffeomorphism of Y restricting to diffeomorphisms $M_1 \rightarrow M'_1$, $M_2 \rightarrow M'_2$, $\Sigma \rightarrow \Sigma'$.

Once you’re happy with all the orientation business, the Casson invariant is obviously invariant under equivalence.

Definition 11 (Stabilization). Let $Y = M_1 \cup_\Sigma M_2$ be a genus g Heegaard splitting. We may use this to construct a genus $g + 1$ Heegaard splitting for Y as follows. Cut an unknotted 1-handle out of M_2 and “give it” to M_1 . As we discussed in class, the effect on M_2 is equivalent to attaching a 1-handle. Note that by “unknotted” we mean that the 1-handle’s core curve, together with M_1 , bounds an embedded disk in M_2 . This disk becomes the cocore for the new

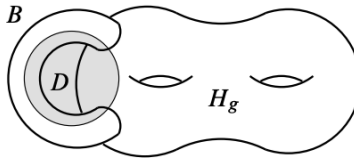


Figure 1: A depiction of stabilization, shamelessly ripped directly from [Sav12].

handle of M_2 . The resulting genus $g + 1$ splitting is called a *stabilization* of $Y = M_1 \cup_{\Sigma} M_2$. Figure 1 provides an illustration of stabilization.

Definition 12 (Stable equivalence). Two Heegaard splittings are *stably equivalent* if they become equivalent after applying some number of stabilizations to each one.

Theorem 3. Any two Heegaard splittings of a closed oriented 3-manifold are stably equivalent.

So, invariance of the Casson invariant reduces to the following theorem.

Theorem 4. The Casson invariant is invariant under stabilization.

References

- [Sav12] Nikolai Saveliev. *Lectures on the Topology of 3-Manifolds: An Introduction to the Casson Invariant*. De Gruyter, Berlin, Boston, 2012.